EDMUND RICE COLLEGE • SUPPORTED LEARNING PROGRAM

Topic 9: Data Analysis

Student Learning Booklet & Guided Teacher Notes.
Covers displaying data, averages (mean, median, mode), spread (range), frequency tables, stem-and-leaf plots, and fair samples.

STUDENT NAME	CLASS / GROUP	TEACHER

Mode
The **MOST** frequent score

Median
The **MIDDLE** after ordering

Mean
Fair share: **Sum ÷ Count**

Range
Highest − Lowest

Displaying Data Review

🎯 **Learning Intention:** We are learning to read, count, and compare data from tables, dot plots, and bar graphs.
💡 **Memory Hook:** On a dot plot, every single dot represents **1 person or 1 count**. The tallest tower is the most popular!

TEACHER GUIDED NOTES & CLASS STARTER EXAMPLES:

Displaying Data (Questions 1 – 3)

1 After-School Activities Dot Plot



a. How many chose **Television**? _____

b. How many chose **Video Games**? _____

c. Most popular activity: _____

d. Total students surveyed (count all dots): _____

2 Pocket Money for Chores

Giselle earns **$3** for each chore she completes around the house.

Chore	Count
Made bed	**4**
Fed dog	**2**
Dishes	**2**

a. Total jobs done: 4 + 2 + 2 = _____ jobs

b. Money earned in 1 week: _____ × $3 = **$** _____

c. For **2 weeks** of chores: **$** _____

3 Running Track Times (Johnson's Laps)

Johnson timed 5 laps around the school track in seconds:

Lap 1	Lap 2	Lap 3	Lap 4	Lap 5
70s	72s	65s	52s	60s

a. Which lap was his **fastest** (lowest time)? Lap _____

b. Which lap was his **slowest** (highest time)? Lap _____

c. Difference between Lap 1 (70s) and Lap 4 (52s): 70 − 52 = _____ seconds

Displaying Data (Questions 4 – 6)

4 Class Pet Survey Bar Reading

A class surveyed pets: **Dogs: 8**, **Cats: 6**, **Fish: 3**, **Birds: 1**.

a. Which pet was the most common? _____

b. How many more students have cats than fish? 6 – 3 = _____

c. Total pets counted: 8 + 6 + 3 + 1 = _____ pets

5 Canal Water Heights Table

Water levels were measured over several days:

152 cm	153 cm	154 cm	155 cm
5 days	6 days	9 days	**12 days**

a. Which water height occurred on the most days? _____ cm

b. On how many days was the water level exactly 154 cm? _____ days

c. Total days recorded: 5 + 6 + 9 + 12 = _____ days

6 Height Grouping Check

In a small reading group of 10 students, **5 students** are taller than 145 cm.

What fraction of students are taller than 145 cm? (5 out of 10):

◯ **1/2 (Half)** ◯ **1/4 (One quarter)** ◯ **1/10 (One tenth)**

Calculating Mean and Mode

MODE = "MOST":
The number that appears most often. If every number appears once, there is *no mode*.

MEAN = "FAIR SHARE / AVERAGE":
Step 1: Add all numbers together (Sum).
Step 2: Divide by how many numbers there are.

TEACHER GUIDED NOTES & CLASS STARTER EXAMPLES:

Mean & Mode (Questions 1 – 3)

1 **Guided Example: Goals in 5 Matches:** `2, 1, 2, 4, 1`

Find the Mode:
1 appears twice. 2 appears twice.
Modes: **1** and _____

Find the Mean:
Add: 2 + 1 + 2 + 4 + 1 = **10**
Divide: 10 ÷ 5 = **Mean:** _____

2 **Pets Owned by Friends:** `1, 4, 1, 2, 2`

a. Which numbers appear most often (Modes)? _____ and _____

b. Add all 5 numbers: 1 + 4 + 1 + 2 + 2 = _____

c. Calculate the Mean: _____ ÷ 5 = _____ pets

3 **Basketball Points in 4 Quarters:** `6, 10, 6, 2`

a. What is the Mode score? _____

b. Sum of all 4 scores: 6 + 10 + 6 + 2 = _____

c. Calculate the Mean: _____ ÷ 4 = _____ points

Mean & Mode (Questions 4 – 6)

4 **Ten-Pin Bowling Scores:** `50, 70, 50, 90, 40`

a. What is the Mode score? _____

b. Add all 5 scores: 50 + 70 + 50 + 90 + 40 = _____

c. Calculate the Mean: _____ ÷ 5 = _____

5 **Cricket Runs:** `10, 20, 15, 5, 0`

Ramesh recorded his runs over 5 innings. What was his mean score?

Step 1 (Add): 10 + 20 + 15 + 5 + 0 = _____

Step 2 (Divide): _____ ÷ 5 = **Mean:** _____ runs

6 **Club Ages:** `12, 12, 14, 18`

a. Mode age: _____ years

b. Total of these 4 ages: 12 + 12 + 14 + 18 = _____

c. If another friend aged **24** joins, the new total is: _____ + 24 = _____
Calculate the new Mean for all 5 friends: _____ ÷ 5 = _____ years

Median and Range

MEDIAN = "MIDDLE SCORE":
1. **Always order scores** from smallest to largest first!
2. Cross off pairs from each end to locate the centre.

RANGE = "SPREAD":
Formula: **Highest Score – Lowest Score**
Tells you how spread out the scores are.

TEACHER GUIDED NOTES & CLASS STARTER EXAMPLES:

Median & Range (Questions 1 – 3)

1 Guided Step-by-Step: 7, 2, 9, 3, 5

Step 1 (Order): 2, 3, 5, 7, 9
Step 2 (Cross ends): ~~2~~, 3, 5, 7, ~~9~~
Step 3 (Cross next pair): ~~2~~, ~~3~~, **5**, ~~7~~, ~~9~~

Median = 5
Range = Highest (9) – Lowest (2) = 7

2 Scores: 8, 3, 5, 1, 6

a. Write in order (smallest to largest): _____

b. Middle score (Median): _____

c. Calculate Range: _____ (Highest) – _____ (Lowest) = _____

3 Basketball Team Heights (cm): 155, 180, 160, 175, 180

a. Put in order from shortest to tallest:

____ , ____ , ____ , ____ , ____

b. Median height: _____ cm

c. Range of heights: 180 – 155 = _____ cm

Median & Range (Questions 4 – 6)

4 **Cyclist Lap Times (seconds):** `10, 23, 14, 12, 19`

a. Ordered times: _____

b. Median time: _____ seconds

c. Range: 23 – 10 = _____ seconds

5 **Challenge: Even Set of Scores:** `2, 4, 6, 8`

When there are 4 numbers, two numbers share the middle: **4 and 6**.

What number is halfway between 4 and 6? (Add 4 + 6 = 10, then 10 ÷ 2 = _____)

So the Median is: _____

6 **Cricket Innings:** `2, 17, 22, 35, 54, 86, 98`

These 7 scores are already ordered from smallest to largest!

a. Cross off 3 from each end. What is the **Median**? _____ runs

b. Calculate the **Range**: 98 – 2 = _____ runs

Analysing Data & Outliers

WHAT IS AN OUTLIER? An outlier is an extreme score that is **much higher** or **much lower** than all other numbers in a dataset.

- Outliers pull the **Mean** up or down heavily.
- The **Median** stays safe in the middle and is usually better to use when outliers exist!

TEACHER GUIDED NOTES & CLASS STARTER EXAMPLES:

Outliers & Analysis (Questions 1 – 3)

1 Hourly Wages at a Local Cafe

Staff 1	Staff 2	Staff 3	Staff 4	Boss
$20	$20	$20	$20	$120

a. Which wage is the extreme **outlier**? **$** _____

b. What is the **Mode** (most common wage)? **$** _____

c. Calculate Mean: $200 ÷ 5 = **$** _____ per hour.

d. Did the boss's wage pull the average (mean) up? ☐ **Yes** ☐ **No**

2 Spot the Outlier in Each Set

a. Test marks out of 100: 70, 72, 68, **12**, 74
Outlier: _____ (Is it unusually high or low? _____)

b. Dog weights (kg): 8, 9, 7, 8, **75**
Outlier: _____ (Is it unusually high or low? _____)

3 Data Vocabulary Review

a. The score that appears most frequently is the _____.

b. The middle score after ordering is the _____.

c. Highest score minus lowest score gives the _____.

d. A score far away from all other scores is an _____.

Outliers & Analysis (Questions 4 – 6)

4 **Reasoning Check: 1, 7, 8, 9, 9, 28**

If the high outlier **28** is erased from the list, what happens to the mean?

- ○ **The mean goes down**
- ○ **The mean stays exactly the same**
- ○ **The mean goes up**

5 **Pocket Money: $10, $10, $15, $25**

a. What is the mode? **$** _____

b. If the highest person receives a huge bonus and gets **$100**, which measure changes the most?

○ **The Mean** ○ **The Mode**

6 **Kids in a Playroom: Ages 2, 4, 6**

a. Mean age of the 3 kids: (2 + 4 + 6) ÷ 3 = 12 ÷ 3 = _____ years

b. An adult aged **40** walks into the room. Is 40 an outlier? ○ **Yes** ○ **No**

Analysing Frequency Tables

WHAT IS FREQUENCY? Frequency simply means **how many times** a score happens!
Tally marks help us count quickly without losing track. Five is written with a cross: ||||
Total (f column): Adding the frequency column gives the total number of people surveyed.

TEACHER GUIDED NOTES & CLASS STARTER EXAMPLES:

Frequency Tables (Questions 1 – 3)

1 Number of Siblings in Year 8

Siblings	Tally	Frequency (Count)
0	||	2
1	||||	**4**
2	|||	3
3	|	1
TOTAL:		**10**

a. Mode score: _____

b. Students with zero siblings: _____

c. Highest number of siblings: _____

d. Range (3 – 0): _____

2 Visits to the Doctor Over 3 Months

Visits	Patients
0 visits	**5**
1 visit	**8**
2 visits	**10**
3 visits	**2**

a. How many people visited **2 times**? _____

b. What is the **Mode** (highest frequency)? _____

c. Total patients: 5 + 8 + 10 + 2 = _____

3 Train Delays (Minutes Late)

Minutes late for 6 trains: **0, 0, 1, 2, 2, 2**

a. Complete frequency count: 0 mins = _____ trains, 1 min = _____ train, 2 mins = _____ trains.

b. Mode delay: _____ minutes

Frequency Tables (Questions 4 – 6)

4 School Drama Play Attendance

Tickets sold: **Night 1: 10** | **Night 2: 20** | **Night 3: 30**

a. Total tickets sold: 10 + 20 + 30 = _____

b. Average (mean) tickets per night: _____ ÷ 3 = _____ tickets

5 Movies Watched on Weekend

Watched 1 movie: **4 students** | Watched 2 movies: **6 students**

a. Total students in this group: 4 + 6 = _____

b. What is the mode number of movies? _____

6 Pet Pigeons Owned

Lowest member has **8 pigeons**. Highest member has **20 pigeons**.

Calculate the Range of pigeons: 20 – 8 = _____ pigeons

Stem-and-Leaf & Dot Plots

HOW TO READ A STEM-AND-LEAF PLOT:

Always look at the **KEY** first! Example: `2 | 5 = 25`

- The **Stem** is the tens digit (2 = 20).
- The **Leaf** is the units digit (5 = 5). So 2 | 5 means the number 25.

TEACHER GUIDED NOTES & CLASS STARTER EXAMPLES:

Plots Practice (Questions 1 – 3)

1 Maths Quiz Marks (out of 40)

Stem	Leaf
1	2 8
2	1 5 5
3	0 4

Key: 2 | 1 = 21

Scores: **12, 18, 21, 25, 25, 30, 34**

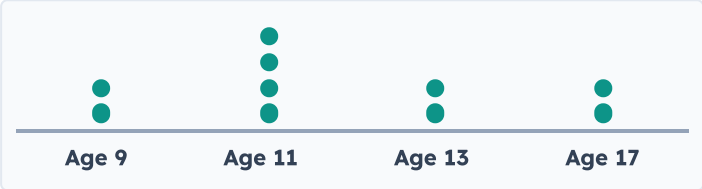
a. Lowest score: _____

b. Highest score: _____

c. Mode score: _____

d. Median (the middle): _____

2 Kate's Cousins' Ages Dot Plot



a. Oldest cousin: _____

b. Youngest cousin: _____

c. Range (17 – 9): _____

d. Mode age: _____

3 Your Turn: Draw a Quick Dot Plot

Draw dots for these goals scored: **1, 2, 2, 3, 2**

Goal: 1	Goal: 2	Goal: 3
[Draw 1 dot]	[Draw 3 dots]	[Draw 1 dot]

Plots Practice (Questions 4 – 6)

4 **Car Speeds: Stem 10, Leaves: 0, 5 (Key: 10 | 5 = 105 km/h)**

a. What speed does 10 | 0 represent? _____ km/h

b. What speed does 10 | 5 represent? _____ km/h

5 **Houses Sold: 5, 5, 6, 7, 7, 7, 8, 9**

a. What is the mode number of houses sold? _____ houses

b. What is the range of houses sold? 9 – 5 = _____ houses

6 **Team Comparison**

The Stars scored between 100 and 140 runs. The Thunder scored between 110 and 130 runs.

Which team had the wider spread (larger range) of scores?

○ **The Stars** ○ **The Thunder**

Samples & Predictions

🏛️ **CENSUS:**
Surveying **EVERY single person** in the whole population (e.g., asking every student in school). Takes much longer!

🎯 **SAMPLE:**
A small, fair group chosen to represent everyone.
⚠️ **Bias:** An unfair sample (e.g. asking only dog breeders about cats).

TEACHER GUIDED NOTES & CLASS STARTER EXAMPLES:

Samples & Predictions (Questions 1 – 3)

1 Spotting Bias (Unfair Surveys)

A researcher wants to find the favourite food of teenagers in town. Which sample is fair?

- ○ **Method A:** Asking 50 teenagers in line outside McDonald's.
- ○ **Method B:** Asking 50 teenagers randomly selected across 5 different high schools.

Why would Method A be biased?

2 Predicting from a Sample (Friendly Numbers)

In a sample of **10 dogs** at a park, **3 dogs** wear red collars.

- In 10 dogs → 3 have red collars.
- In 20 dogs → 6 have red collars.
- In **100 dogs** across town, about how many wear red collars?

Prediction: _____ dogs

3 Soccer Club Fair Sample

There are 3 times as many boys as girls in a junior soccer club. Which sample is fair?

- ○ **Surveying only boys**
- ○ **4 boys and 4 girls**
- ○ **12 boys and 4 girls (3 to 1 ratio)**

Samples & Predictions (Questions 4 – 6)

4 Penguin Colony Sample

In a sample of **10 penguins**, exactly **2 penguins** have belly spots.

If there are **100 penguins** on the island, how many would you predict have spots?
Prediction: _____ penguins

5 Tasmanian Devil Health Check

A vet tests 10 Tasmanian devils and finds 7 are healthy.

In a forest of **100 devils**, about how many are predicted to be healthy? _____

6 Census or Sample?

Decide if each statement describes a **Census** (whole group) or a **Sample** (small fair group):

a. Asking every student in Year 8: _____

b. Asking 20 randomly chosen students from Year 8: _____

Frayer Models: Key Concepts

VOCABULARY CONCEPT: THE MEAN	
1. Definition The fair-share average found by adding all scores and dividing by the total count.	**2. Key Facts** • Uses every single score. • Can be pulled up or down by an outlier!
3. Example Scores: 2, 4, 6 Add: 2 + 4 + 6 = 12 Divide: 12 ÷ 3 = **4**	**4. Non-Example** Just picking the middle number without adding. (That is the *Median*, not the Mean!)

VOCABULARY CONCEPT: THE MEDIAN	
1. Definition The _____ score when numbers are ordered from smallest to largest.	**2. Key Facts** You **must** put scores in order first before crossing off ends!
3. Example Scores in order: 1, 3, [5], 7, 9 The median is: _____	**4. Non-Example** Picking 9 from 2, 9, 3 because 9 is written in the middle without sorting!

Success Criteria Checklist

I can...	Need Help 😒	Getting There 🤔	I Got This! 😃
Read frequencies from a dot plot and table	☐	☐	☐
Find the **Mode** (most common score)	☐	☐	☐
Calculate the **Mean** (Sum ÷ Count)	☐	☐	☐
Find the **Median** by ordering and crossing ends	☐	☐	☐
Calculate the **Range** (Highest – Lowest)	☐	☐	☐
Identify an **outlier** that is far from other scores	☐	☐	☐

TEACHER / LEARNING SUPPORT OFFICER FEEDBACK: